TECHNICAL DILIGENCE REVIEW - RECORD 017

Helioscope Robotics, Inc. - diligence file reference TECHNICAL_DILIGENCE-017

1. Instruction, scope and the evidence relied on

As instructed, Brandywine Technology Partners confined TDR-2026-13 to the service architecture, ownership boundaries and dependency structure of the fleet platform; three adjacent questions raised during the work are listed as follow-ups rather than answered here. This report addresses the service architecture, ownership boundaries and dependency structure of the fleet platform. It does not address commercial terms, financial reporting or employment matters, which sit with other workstreams. Report TDR-2026-13 carries one structural limitation: the company's own metrics are the only source for several objectives, and the reviewer could not build an independent measurement inside the engagement. The reviewer notes that this report was prepared while three remediation programmes were in flight, so several positions will have moved by completion. The service architecture, ownership boundaries and dependency structure of the fleet platform produced more evidence than the present report reproduces, and the working papers hold the remainder.

The working method behind this report was to start from the incident record and work back to the design, rather than the other way round. Method in the report at TDR-2026-13 was three-fold: read the design record, interview the owning engineers, and verify the claim against a running system. 3 items in TDR-2026-13 sit in a declined state, and the reviewer regards one of them as wrongly declined. R. Ostrowski asked in every declined case who would revisit the decision and when, and the report at TDR-2026-13 records the answer even where there was none. R. Ostrowski has kept the working papers behind the report at TDR-2026-13 in a form that a second reviewer could re-perform.

The population for the present report was the service inventory as at 2026-04-13; 10 entries were examined and the rest were reviewed only at inventory level. 4 sites contributed evidence to the present report, and one further site was in acceptance testing and was excluded. Not every finding in this report is accepted by management: 3 are declined, and the declinatures are reproduced rather than argued away. 3 declined items are carried in report TDR-2026-13, of which two lack any documented compensating control. Brandywine Technology Partners raised each finding with the owning engineer before it was written up, and the responses are recorded.

TDR-2026-13 reports two findings the company disputes; the dispute is recorded next to the finding rather than resolved silently. R. Ostrowski treats a backlog ticket as evidence of awareness and a merged change with a test as evidence of remediation, and this report keeps the two apart. Reporting under this report should distinguish a target that was met from a target that was lowered, which the current pack does not. Report TDR-2026-13 recommends that the remediation commitment register be reported to the board quarterly with evidence attached, not just status. The evidence behind TDR-2026-13 is retained by Brandywine Technology Partners and is available to Ardent Peak on request.

R. Ostrowski could not verify one class of claim in report TDR-2026-13, namely the behaviour of components deployed only at the two sites not visited. Nothing in this report rests on a test the reviewer designed; the tests were the company's, and the reviewer read the results and the method. TDR-2026-13 sampled 10 services from an inventory of 34, weighted towards those on the critical control path. 10 services were examined in detail for TDR-2026-13, chosen for their position in the dependency map rather than at random. The service architecture, ownership boundaries and dependency structure of the fleet platform is reported here at the level of detail Ardent Peak asked for and no deeper.

The interview programme behind report TDR-2026-13 covered 8 teams and produced 17 follow-up requests, of which most were satisfied. Interview evidence in the report at TDR-2026-13 came from 16 sessions across 8 teams, each minuted and circulated for correction. 5 matters remain open at the date of report TDR-2026-13, and none of them is closed by the passage of time alone. R. Ostrowski has dated each open item in report TDR-2026-13 so that a reader can see which have been open longest rather than only how many there are. The reviewer's tone in report TDR-2026-13 is deliberately flat; a finding is not made more serious by the words around it.

Schedule E - Index of evidence relied on, with the use made of each item (TDR-2026-13)

Evidence ref	Document or artefact	Type	Date	Provided by	Relied on for
EV-17-001	Event Bus: test report	test report	2025-07-18	the safety controller vendor	supported a capacity conclusion

Evidence ref	Document or artefact	Type	Date	Provided by	Relied on for
EV-17-002	release packaging and signing: runbook	runbook	2025-08-30	Helioscope engineering	evidenced an approval
EV-17-003	Release Orchestrator: interview minute	interview minute	2026-10-01	the safety controller vendor	supported a coverage figure
EV-17-004	dependency and licence scan: measurement extract	measurement extract	2025-10-02	the sensor vendor	established provenance
EV-17-005	Payload Scale Interface: incident post-mortem	incident post-mortem	2026-08-23	the safety controller vendor	supported a coverage figure
EV-17-006	container image vulnerability scan: release record	release record	2026-04-25	Helioscope security	supported a coverage figure
EV-17-007	release packaging and signing: policy document	policy document	2026-06-26	Helioscope safety	established a dependency
EV-17-008	Order Broker: policy document	policy document	2026-11-10	the safety controller vendor	supported a coverage figure
EV-17-009	Wheel Odometry Filter: runbook	runbook	2025-05-16	Helioscope security	supported a coverage figure
EV-17-010	lint and format: incident post-mortem	incident post-mortem	2026-07-18	the safety controller vendor	supported a coverage figure
EV-17-011	Lidar Driver: audit output	audit output	2025-09-03	Helioscope field reliability	established provenance
EV-17-012	Payload Scale Interface: measurement extract	measurement extract	2026-06-05	the sensor vendor	established a dependency
EV-17-013	build, robot firmware images: release record	release record	2025-06-22	the safety controller vendor	evidenced an approval
EV-17-014	Wheel Odometry Filter: incident post-mortem	incident post-mortem	2026-01-01	Meridian Assurance	supported a coverage figure
EV-17-015	Simulation Harness: audit output	audit output	2026-05-17	Helioscope safety	evidenced a remediation
EV-17-016	Release Orchestrator: measurement extract	measurement extract	2025-11-03	Helioscope security	evidenced an approval
EV-17-017	dependency and licence scan: release record	release record	2026-03-13	the safety controller vendor	established a dependency
EV-17-018	container image vulnerability scan: architecture note	architecture note	2026-09-02	Helioscope field reliability	evidenced an approval
EV-17-019	Simulation Harness: release record	release record	2026-07-14	Helioscope security	corroborated an incident narrative
EV-17-020	build, cloud container images: design record	design record	2026-01-25	Helioscope field reliability	established a dependency
EV-17-021	OTA Updater: incident post-mortem	incident post-mortem	2026-03-07	the sensor vendor	verified a stated objective
EV-17-022	Speed and Separation Monitor: audit output	audit output	2026-09-21	Helioscope engineering	established provenance
EV-17-023	Order Broker: policy document	policy document	2025-08-23	the safety controller vendor	evidenced a remediation
EV-17-024	Event Bus: vendor statement	vendor statement	2025-11-12	Helioscope security	established provenance
EV-17-025	Payload Scale Interface: release record	release record	2026-06-12	Helioscope security	evidenced a remediation
EV-17-026	OTA Updater: audit output	audit output	2025-08-13	Helioscope field reliability	corroborated an incident narrative
EV-17-027	performance smoke, cloud: configuration export	configuration export	2025-05-10	the sensor vendor	established provenance
EV-17-028	Charge Scheduler: architecture note	architecture note	2025-10-03	the safety controller vendor	supported a coverage figure
EV-17-029	unit tests, robot software: test report	test report	2025-05-17	the sensor vendor	supported a coverage figure
EV-17-030	Config Distribution: interview minute	interview minute	2026-07-14	Helioscope engineering	verified a stated objective

Evidence ref	Document or artefact	Type	Date	Provided by	Relied on for
EV-17-031	Robot Shadow Store: incident post-mortem	incident post-mortem	2026-04-16	Helioscope safety	verified a stated objective
EV-17-032	secret scan: release record	release record	2026-11-01	Helioscope engineering	corroborated an incident narrative
EV-17-033	Robot Shadow Store: interview minute	interview minute	2026-10-08	the sensor vendor	supported a coverage figure
EV-17-034	Wheel Odometry Filter: policy document	policy document	2025-08-09	the safety controller vendor	corroborated an incident narrative
EV-17-035	lint and format: policy document	policy document	2026-07-26	the sensor vendor	corroborated an incident narrative
EV-17-036	Telemetry Ingest: test report	test report	2026-03-30	Meridian Assurance	supported a coverage figure

1. Instruction, scope and the evidence relied on - part 2 of 2

Where the 1. Instruction, scope and the evidence relied on was interrupted by the page, it picks up again below, and the numbering is continuous throughout the schedule. The break is editorial.

No basis of compilation changes at this break; whatever the 1. Instruction, scope and the evidence relied on recorded about scope and method at its head continues to apply. Nothing is restated from the earlier part.

Schedule E - Index of evidence relied on, with the use made of each item (TDR-2026-13) (continued, part 2 of 2)

Evidence ref	Document or artefact	Type	Date	Provided by	Relied on for
EV-17-037	Costmap Fusion: interview minute	interview minute	2026-09-21	Helioscope safety	supported a capacity conclusion
EV-17-038	Event Bus: interview minute	interview minute	2025-08-27	Helioscope safety	established provenance
EV-17-039	Order Broker: configuration export	configuration export	2026-01-14	Meridian Assurance	evidenced a remediation
EV-17-040	Traffic Manager: vendor statement	vendor statement	2025-07-24	the sensor vendor	supported a capacity conclusion
EV-17-041	simulation suite, nightly: configuration export	configuration export	2025-10-14	Helioscope engineering	verified a stated objective
EV-17-042	release packaging and signing: incident post-mortem	incident post-mortem	2026-08-19	Helioscope safety	evidenced an approval
EV-17-043	integration tests, cloud: audit output	audit output	2025-09-20	Helioscope safety	evidenced an approval
EV-17-044	Payload Scale Interface: configuration export	configuration export	2026-11-02	Redwater Security	supported a capacity conclusion
EV-17-045	Telemetry Ingest: vendor statement	vendor statement	2026-06-08	Meridian Assurance	supported a capacity conclusion
EV-17-046	Gripper Controller: measurement extract	measurement extract	2025-07-04	Helioscope engineering	evidenced an approval
EV-17-047	Site Provisioning Service: runbook	runbook	2025-11-01	Helioscope engineering	corroborated an incident narrative
EV-17-048	integration tests, cloud: configuration export	configuration export	2025-12-09	the sensor vendor	evidenced a remediation
EV-17-049	static analysis, robot software: measurement extract	measurement extract	2025-12-26	Helioscope engineering	verified a stated objective
EV-17-050	Payload Scale Interface: measurement extract	measurement extract	2025-09-08	Helioscope engineering	supported a capacity conclusion
EV-17-051	Map Authoring Studio: interview minute	interview minute	2025-10-12	Helioscope security	corroborated an incident narrative
EV-17-052	hardware-in-the-loop, bench: design record	design record	2025-08-20	Helioscope engineering	evidenced a remediation
EV-17-053	Gripper Controller: audit output	audit output	2025-11-13	Meridian Assurance	supported a coverage figure
EV-17-054	Edge Gateway Agent: policy document	policy document	2026-07-17	Helioscope engineering	evidenced a remediation

Evidence ref	Document or artefact	Type	Date	Provided by	Relied on for
EV-17-055	performance smoke, cloud: release record	release record	2026-01-09	Redwater Security	evidenced a remediation
EV-17-056	Costmap Fusion: incident post-mortem	incident post-mortem	2026-10-11	Redwater Security	established a dependency
EV-17-057	Order Broker: audit output	audit output	2026-07-17	Meridian Assurance	established a dependency
EV-17-058	Config Distribution: release record	release record	2026-08-13	Helioscope engineering	established a dependency
EV-17-059	release packaging and signing: audit output	audit output	2025-11-08	Meridian Assurance	established provenance
EV-17-060	Identity and Authorization: design record	design record	2025-05-02	Helioscope field reliability	evidenced an approval
EV-17-061	release packaging and signing: architecture note	architecture note	2026-05-11	Meridian Assurance	established a dependency
EV-17-062	WMS Connector: measurement extract	measurement extract	2025-12-04	the safety controller vendor	supported a coverage figure
EV-17-063	performance smoke, cloud: interview minute	interview minute	2025-10-29	the sensor vendor	evidenced a remediation
EV-17-064	Motion Planner: measurement extract	measurement extract	2026-09-11	Redwater Security	evidenced a remediation
EV-17-065	Edge Gateway Agent: audit output	audit output	2026-02-11	Helioscope safety	corroborated an incident narrative
EV-17-066	Traffic Manager: design record	design record	2026-06-16	Helioscope field reliability	verified a stated objective
EV-17-067	static analysis, robot software: incident post-mortem	incident post-mortem	2025-12-02	Helioscope security	evidenced a remediation
EV-17-068	Gripper Controller: incident post-mortem	incident post-mortem	2026-05-24	Helioscope safety	established provenance
EV-17-069	static analysis, robot software: architecture note	architecture note	2025-09-11	Meridian Assurance	established provenance
EV-17-070	static analysis, robot software: design record	design record	2025-09-19	the sensor vendor	established a dependency
EV-17-071	Site Provisioning Service: configuration export	configuration export	2025-11-29	Meridian Assurance	established provenance
EV-17-072	static analysis, robot software: interview minute	interview minute	2026-02-14	Helioscope field reliability	supported a capacity conclusion

2. Costmap Fusion and the services on the control path

21 defects remain open against Costmap Fusion; the one the reviewer pressed on is that a stale lidar frame is retained for one extra cycle while a driver reconnects. Costmap Fusion is owned by the Perception team, with Daniel Okafor named as the engineering owner of record. Placement of Costmap Fusion is edge, co-scheduled with the planner on the compute module, which the reviewer tested by tracing a request end to end. On paper Costmap Fusion releases monthly, batched with perception model updates, and the observed practice did not diverge from that. End-of-life review of Costmap Fusion found a vendored geometry library whose upstream branch left support in 2024. Costmap Fusion sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

Reading the dependency map from Costmap Fusion outward gives the lidar driver, the depth camera bridge and the wheel odometry filter on the input side and the motion planner and the speed and separation monitor on the output side. Evidence examined for Costmap Fusion comprised the fusion design note, sensor timing captures and the reviewer's own bench replay. Costmap Fusion reports 58 per cent line coverage, measured by the same instrumentation the pipeline uses. Where Costmap Fusion is concerned, the commitment is that fused occupancy grid published at 20 Hz with sensor-to-grid latency under 60 ms; the reviewer traced the metric to its source. Costmap Fusion was assessed against the objective and the defect record, and the reviewer could corroborate the availability claim but not the latency claim, which rests on a single dashboard. Costmap Fusion is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

The interface Gripper Controller publishes is versioned, and the order execution path and the safety controller fault input were checked for use of any undocumented field. End-of-life review of Gripper Controller found none; the coprocessor part is current production. Gripper Controller is written in C++17, executes on on-robot Linux with a firmware coprocessor, and keeps its state in a local parameter store. Reading the dependency map from Gripper Controller outward gives the payload scale interface and the perception grasp proposals on the input side and the order execution path and the safety controller fault input on the output side. The assessment of Gripper Controller is not unqualified: the position is stronger than the coverage figure alone would suggest, because the bench evidence is unusually good here. Gripper Controller is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Robot Firmware holds Gripper Controller, and the ownership entry naming Ines Duarte matches the on-call rotation. 41 per cent line coverage is reported for Gripper Controller, and the reviewer re-ran the measurement on a clean checkout. Deployment of Gripper Controller is edge, only on the two robot variants fitted with a manipulator, which sets the blast radius of any defect in it. Against Gripper Controller the backlog runs to 15 items, and it includes a known behaviour whereby force limiting relies on a calibration that is only re-run when a gripper is replaced. Underlying Gripper Controller in this report are the gripper control specification, calibration records for eleven units and a force bench test. Gripper Controller is included because a reader of TDR-2026-13 should be able to disagree with the inference without repeating the work.

The Payload Scale Interface codebase is C++17 throughout, deployed onto on-robot Linux, user-space driver, and its durable record is a local calibration record. The gripper controller and the order execution path rely on Payload Scale Interface, which in turn relies on the load cell hardware. The reviewer's judgement on Payload Scale Interface is that no verified defect was identified, and two plausible concerns are recorded below at low evidence strength. Ownership of Payload Scale Interface sits with Robot Firmware; Ines Duarte signs off changes to it. Test coverage on Payload Scale Interface is 36 per cent line coverage and the measurement basis was confirmed with Ines Duarte. Payload Scale Interface is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

The manifests place Payload Scale Interface edge, only on robots fitted with the weighing deck. Payload Scale Interface was tested against the dependency ageing register, which records none identified. Payload Scale Interface carries a single measurable objective, namely that payload mass is reported within 2 per cent of the reference load. Release cadence for Payload Scale Interface is as needed, and the release records bear that out. Payload Scale Interface carries 11 open defects, among them a case where the calibration record is stored locally and is lost when the compute module is swapped. Payload Scale Interface would be a candidate for the first post-close engineering review under any reasonable plan.

For Fleet Allocator the reviewer records that management's own assessment is broadly supported, with the coverage gap treated as a genuine finding rather than a presentational one. Fleet Services publishes one objective for Fleet Allocator: task assignment inside 900 ms of order arrival at p95. Inputs to Fleet Allocator come from the order broker and the robot shadow store, and its consumers are the traffic manager, the charge scheduler and the fleet health monitor. The open defect backlog for Fleet Allocator holds 9 items, the most consequential being that assignment starvation for robots parked at chargers below the battery reserve threshold. Fleet Allocator sits cloud, one regional deployment serving all sites in that region, and no alternative placement is currently supported. Fleet Allocator was raised again with the owning engineer after the initial interview, and the position did not change.

The interface Fleet Allocator publishes is versioned, and the traffic manager, the charge scheduler and the fleet health monitor were checked for use of any undocumented field. Implementation of Fleet Allocator rests on Go 1.22; the process runs on cloud, containers on a managed Kubernetes cluster and persists to PostgreSQL 14 with a synchronous standby. Fleet Allocator rests on the allocator service contract, two design reviews and a week of assignment audit logs so far as this review is concerned. Release cadence for Fleet Allocator is weekly, behind a change approval, and the release records bear that out. Test coverage on Fleet Allocator is 82 per cent line coverage and the measurement basis was confirmed with Marta Kellerman. Fleet Allocator was checked against the company's own register, which had not been updated since the change landed.

Schedule A - Service inventory, ownership and objective attainment (TDR-2026-13)

Service	Language	Owning team	Coverage	Service-level objective	Upstream dependencies	End-of-life exposure
Motion Planner	Rust 1.79	Autonomy Core	71%	plan cycle completes inside 40 ms at the 99th percentile	the costmap fusion pipeline and the localization engine	none; the toolchain is pinned to a supported compiler release
Costmap Fusion	C++17	Perception	58%	fused occupancy grid published at 20 Hz with sensor-to-grid latency under 60 ms	the lidar driver, the depth camera bridge and the wheel odometry filter	a vendored geometry library whose upstream branch left support in 2024
Fleet Allocator	Go 1.22	Fleet Services	82%	task assignment inside 900 ms of order arrival at p95	the order broker and the robot shadow store	none identified in the reviewed manifests
Traffic Manager	Go 1.22	Fleet Services	74%	corridor deadlock detected and broken inside two seconds	the fleet allocator and the site map service	Redis 6 client library still vendored in one code path
Charge Scheduler	Python 3.11	Fleet Services	63%	fleet state of charge stays above 35 per cent through the peak shift at every site	the battery management service and the site shift calendar	none; the dependency set was refreshed in the 2026 platform upgrade
Safety Controller	C, written to a constrained coding subset	Safety Engineering	91%	a validated protective-stop trigger de-energises the drive inside 120 ms	the E-stop chain, the speed and separation monitor and the drive current sensors	none; the supplier's support commitment runs to 2031
E-Stop Chain	VHDL	Safety Engineering	witnessed test	drive power is removed within 250 ms of button actuation on every tested unit	the operator stop buttons and the perimeter interlocks	none; the relay parts are current production
Lidar Driver	C++17	Robot Firmware	44%	no more than one dropped scan per ten thousand at nominal spin rate	the sensor hardware and the time synchronisation daemon	the vendor's software development kit is two major versions behind current
Localization Engine	C++17	Autonomy Core	66%	pose error stays under 25 mm in mapped aisles at survey speed	the lidar driver, the wheel odometry filter and the map distribution channel	an unmaintained scan-matching library is still linked in the fallback path
WMS Connector	Java 21	Integrations	69%	order acknowledgement returned to the warehouse system inside 1.5 s at p95	the customer warehouse management systems and the identity service	one customer integration still runs on a connector build from 2024
Order Broker	Go 1.22	Fleet Services	78%	no accepted order is lost, measured by journal reconciliation each night	the WMS connector and the site configuration service	none identified
Telemetry Ingest	Go 1.22	Data and Telemetry	61%	telemetry visible in the operator console inside 30 s of emission at p95	the edge gateway agent and the event bus	the stream processor runs a release that leaves community support in late 2026
OTA Updater	Rust 1.79 on the robot, Go 1.22 in the cloud	Platform Infrastructure	76%	an update is either fully applied or fully rolled back, with no mixed-version robot	the release orchestrator and the artefact signing service	none; artefact signing was moved to the current key format in 2025
Identity and	Go 1.22	Security	84%	token issuance	the customer	none identified in

Service	Language	Owning team	Coverage	Service-level objective	Upstream dependencies	End-of-life exposure
Authorization		Engineering		inside 200 ms at p95 and no unauthenticated write path	directory federations and the secret store	the reviewed manifests
Remote Operations Console	TypeScript	Applications	52%	operator input reaches the robot inside 350 ms at p95 on a supported network	the identity service, the telemetry ingest path and the video snippet service	a charting library that no longer receives security updates
Simulation Harness	Python 3.11	Developer Productivity	47%	the nightly scenario suite completes inside four hours	the planner and localization builds under test	the physics core depends on a library branch that stopped receiving fixes in 2023
Hardware-in-the-Loop Rig	Python 3.11 test harness around production firmware	Robot Firmware	39%	every firmware candidate runs the full bench suite before ring one	the firmware candidate artefacts and the bench sensor stimulators	the rig controller runs an operating system release that left vendor support in 2025

2. Costmap Fusion and the services on the control path - part 2 of 2

A further extract of the 2. Costmap Fusion and the services on the control path appears here, taken in order from the point at which the previous part ended. Nothing is restated from the earlier part.

Definitions given once for the 2. Costmap Fusion and the services on the control path are relied on again rather than restated, the compilation basis being identical. The break is editorial.

Schedule A - Service inventory, ownership and objective attainment (TDR-2026-13) (continued, part 2 of 2)

Service	Language	Owning team	Coverage	Service-level objective	Upstream dependencies	End-of-life exposure
Fleet Health Monitor	Python 3.11	Field Reliability	57%	a fleet-affecting condition raises an alert inside three minutes of first symptom	the telemetry ingest path and the robot shadow store	none identified
Battery Management Service	C on the pack controller, Python 3.11 in the cloud	Robot Firmware	54%	pack cell imbalance is reported before it reaches the derating threshold	the pack sensors and the charge scheduler	one pack revision uses a cell supplier that has since discontinued the part
Gripper Controller	C++17	Robot Firmware	41%	a grasp completes or reports failure inside 1.2 s	the payload scale interface and the perception grasp proposals	none; the coprocessor part is current production
Map Authoring Studio	TypeScript	Applications	49%	a published map revision reaches every robot at the site inside fifteen minutes	the site survey imports and the identity service	a geometry rendering library one major version behind its supported line
Site Provisioning Service	Go 1.22	Platform Infrastructure	64%	a new site reaches a working configuration inside one working day	the identity service and the customer network intake records	none identified
Event Bus	operated open-source broker, no in-house code	Platform Infrastructure	witnessed test	publish acknowledgement inside 25 ms at p95 and no partition without a leader	every publishing service	the broker minor release in production leaves the vendor's support window in 2027
Config Distribution	Rust 1.79 agent	Platform Infrastructure	72%	a configuration change reaches every online robot inside five minutes	the site provisioning service and the artefact signing service	none; the bundle format was revised in 2025
Robot Shadow	Go 1.22	Fleet Services	68%	shadow state is no	the telemetry	none identified

Service	Language	Owning team	Coverage	Service-level objective	Upstream dependencies	End-of-life exposure
Store				more than two seconds behind the robot at p95	ingest path and the edge gateway agent	
Analytics Warehouse Loader	Python 3.11	Data and Telemetry	45%	the daily load completes before 06:00 local time at every region	the telemetry ingest path, the order broker and the billing usage meter	one transformation still runs on a deprecated warehouse function
Billing Usage Meter	Go 1.22	Data and Telemetry	79%	metered usage reconciles to the order journal within 0.5 per cent each month	the order broker and the WMS connector	none identified
Alarm Router	Python 3.11	Field Reliability	51%	a dispatched alarm reaches an on-call responder inside 90 s at p95	the fleet health monitor and the on-call rotation source	the paging provider's older application interface is still in use
Video Snippet Service	Go 1.22	Applications	43%	a requested snippet is available to the operator inside 45 s	the on-robot camera buffer and the identity service	a transcoder build with known but low-severity upstream advisories
Speed and Separation Monitor	C++17	Safety Engineering	87%	commanded speed is reduced before the separation distance falls below the protective distance	the costmap fusion pipeline and the safety-rated scanner inputs	none; the sensing hardware is current production
Wheel Odometry Filter	C++17	Robot Firmware	62%	odometry drift stays under 1.5 per cent of distance travelled on a clean floor	the wheel encoders and the inertial measurement unit	none identified
Payload Scale Interface	C++17	Robot Firmware	36%	payload mass is reported within 2 per cent of the reference load	the load cell hardware	none identified
Edge Gateway Agent	Rust 1.79	Platform Infrastructure	74%	no telemetry is lost for an outage shorter than four hours	the on-robot services that emit telemetry	none; the transport library was updated in early 2026
Release Orchestrator	Go 1.22	Developer Productivity	55%	no artefact reaches production without a recorded approval and a passing gate set	the build system, the simulation harness gate and the identity service	none identified

3. Deployment topology and failure domains, beginning with Battery Management Service

Taking the evidence together, Battery Management Service the component is adequate today and carries a supportability liability that is not on the funded plan. The manifests place Battery Management Service split edge and cloud. Battery Management Service shows 54 per cent line coverage in the pipeline, and the figure is respectable but excludes the fallback path, which is where the known defect sits. Of the 10 open items against Battery Management Service, the reviewer treats one as material: state-of-health estimates drift on packs that spend most of their life on opportunity charge. Release cadence for Battery Management Service is pack firmware twice a year, the cloud service monthly, and the release records bear that out. Battery Management Service is the clearest instance in TDR-2026-13 of engineering practice being better than the documentation suggests.

On end-of-life exposure, Battery Management Service shows one pack revision uses a cell supplier that has since discontinued the part. Battery Management Service carries a single measurable objective, namely that pack cell imbalance

is reported before it reaches the derating threshold. Pack non-volatile memory and a cloud time-series store is the system of record for Battery Management Service, a C on the pack controller, Python 3.11 in the cloud component running on split: pack firmware and a cloud analytics service. Battery Management Service was checked for shared-database coupling, and the ownership boundary held. Robot Firmware holds Battery Management Service, and the ownership entry naming Peter Halvorsen matches the on-call rotation. Battery Management Service has no single accountable owner, and that is the finding the reviewer would press hardest.

Config Distribution owns its own data and exposes it only through a versioned interface, which the reviewer read. Implementation of Config Distribution rests on Rust 1.79 agent with a Go 1.22 control service; the process runs on split: an on-robot agent and a cloud service and persists to content-addressed object storage with a PostgreSQL 14 index. Of the 6 open items against Config Distribution, the reviewer treats one as material: a robot that is offline at publication applies the change without re-validating preconditions. What the reviewer had for Config Distribution was the bundle format specification, distribution latency measurements and an offline apply test; the conclusions below do not reach past it. Config Distribution sits at 72 per cent line coverage; the trend across the reviewed period was up by seven points. Config Distribution is reported at the evidence strength available and has not been promoted to fit the narrative.

Deployment of Config Distribution is split edge and cloud, with a signed bundle format, which sets the blast radius of any defect in it. Platform Infrastructure publishes one objective for Config Distribution: a configuration change reaches every online robot inside five minutes. On paper Config Distribution releases weekly, and the observed practice did not diverge from that. Taking the evidence together, Config Distribution the reviewer accepts the engineering position while noting that the verifying test does not exist. Sam Oyelaran of Platform Infrastructure maintains Config Distribution, and the service inventory entry was current at the review date. Config Distribution would move from a concern to a verified defect if one further measurement were supplied.

Site Provisioning Service is owned by the Platform Infrastructure team, with Marcus Duplessis named as the engineering owner of record. On end-of-life exposure, Site Provisioning Service shows none identified. Implementation of Site Provisioning Service rests on Go 1.22; the process runs on cloud, containers on a managed Kubernetes cluster and persists to PostgreSQL 14 with a configuration history table. Site Provisioning Service is deployed cloud, global. On Site Provisioning Service the conclusion is qualified: the position is stronger than the coverage figure alone would suggest, because the bench evidence is unusually good here. Site Provisioning Service was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

5 defects remain open against Site Provisioning Service; the one the reviewer pressed on is that site templates diverged after two customers required bespoke network settings. For Site Provisioning Service the stated target is that a new site reaches a working configuration inside one working day, expressed against production traffic rather than a benchmark. Site Provisioning Service reports 64 per cent line coverage, measured by the same instrumentation the pipeline uses. Documents and data inspected for Site Provisioning Service were the provisioning runbook, template diffs for four sites and two go-live records. Changes to Site Provisioning Service reach production fortnightly. Site Provisioning Service rests on a single source, and the reviewer has said so rather than implying corroboration.

3 items sit in the Safety Controller backlog; the entry that bears on the investment case is that the fault log wraps after four thousand entries and the oldest records are discarded. Where Safety Controller is concerned, the commitment is that a validated protective-stop trigger de-energises the drive inside 120 ms; the reviewer traced the metric to its source. Data ownership is clear for Safety Controller: it writes a non-volatile fault log on the controller and no other service writes there. The assessment of Safety Controller is not unqualified: the component is adequate today and carries a supportability liability that is not on the funded plan. Safety Controller was built in C, written to a constrained coding subset and has not been rewritten; it still runs on a safety-rated controller module supplied by a third party against a non-volatile fault log on the controller. Safety Controller was reviewed against the position twelve months earlier, and the direction of travel is favourable.

Automated test coverage on Safety Controller stands at 91 per cent line coverage. In production Safety Controller is edge, dual-channel, with cross-checking between channels, unchanged since the last architecture review. Safety Controller has a single named owner, Bettina Vogel of Safety Engineering, which is not true of every service reviewed. Safety Controller was examined for unsupported components; none; the supplier's support commitment runs to 2031. The evidence base for Safety Controller was the supplier assessment report, the internal safety case and eight bench trigger

tests. Safety Controller is one of the entries where TDR-2026-13 relies on the company's own measurement rather than an independent one.

For Analytics Warehouse Loader the stated target is that the daily load completes before 06:00 local time at every region, expressed against production traffic rather than a benchmark. Analytics Warehouse Loader depends on the telemetry ingest path, the order broker and the billing usage meter and is depended on by the customer reporting views and the internal metrics layer. Analytics Warehouse Loader reports 45 per cent line coverage, measured by the same instrumentation the pipeline uses. Underlying Analytics Warehouse Loader in this report are the load schedule, partition completeness checks and two months of load audit records. Changes to Analytics Warehouse Loader reach production weekly. Analytics Warehouse Loader was compared with the equivalent position at two other components, and it is neither the best nor the worst.

Analytics Warehouse Loader does not share a schema with any consumer; every read by the customer reporting views and the internal metrics layer goes through the published interface. The open defect backlog for Analytics Warehouse Loader holds 20 items, the most consequential being that a late-arriving telemetry partition is silently excluded rather than reprocessed. Implementation of Analytics Warehouse Loader rests on Python 3.11 with SQL transformations; the process runs on cloud, a scheduled batch runner and persists to a columnar warehouse with daily partitions. In production Analytics Warehouse Loader is cloud, one loader per region writing to a shared warehouse, unchanged since the last architecture review. End-of-life review of Analytics Warehouse Loader found one transformation still runs on a deprecated warehouse function. Analytics Warehouse Loader is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

Schedule B - Deployment topology and failure domain by service (TDR-2026-13)

Service	Deployment topology	Release cadence	Instances per region	Failure domain	Engineering owner
Motion Planner	edge, one instance per robot with no cloud fallback	every three weeks, promoted ring by ring	6	one robot	Priya Raghunathan
Costmap Fusion	edge, co-scheduled with the planner on the compute module	monthly, batched with perception model updates	5	one robot	Daniel Okafor
Fleet Allocator	cloud, one regional deployment serving all sites in that region	weekly, behind a change approval	3	one region	Marta Kellerman
Traffic Manager	cloud, regional, with a degraded on-robot yield behaviour if the link drops	weekly, paired with the allocator release	3	one region	Owen Brantley
Charge Scheduler	cloud, regional, with a conservative local fallback policy on the robot	fortnightly, without a formal approval gate until March 2026	4	one region	Nadia Semenova
Safety Controller	edge, dual-channel, with cross-checking between channels	twice a year, with full revalidation of the safety case	4	one robot	Bettina Vogel
E-Stop Chain	edge, hardwired, independent of the compute module	changed only by an engineering change order	3	one robot	Theo Bassett
Lidar Driver	edge, one process per sensor	as needed, historically twice a year	2	one robot	Yuki Harada
Localization Engine	edge, with map distribution from the cloud	every three weeks with the planner	3	one robot	Rahul Menon
WMS Connector	cloud, one deployment per customer tenant	every two weeks, per tenant on request	2	one region	Tomas Villareal
Order Broker	cloud, regional	weekly	2	one region	Grace Lindqvist
Telemetry Ingest	cloud, one ingest endpoint per region with an edge buffer on the robot	weekly, with schema changes gated on a compatibility check	2	one robot	Farida Cisse
OTA Updater	split edge and cloud, with staged rings and a per-site pause	every three weeks, following the autonomy release	3	one robot	Erik Palmgren

Service	Deployment topology	Release cadence	Instances per region	Failure domain	Engineering owner
	control				
Identity and Authorization	cloud, global control plane with regional caches	weekly, with a security review on every change	1	one region	Sonia Krall
Remote Operations Console	cloud, with a thin session relay in each region	twice a week	5	one region	Anjali Sekhon
Simulation Harness	cloud batch, with a local mode for developers	continuous, from the main branch	3	one region	Colin Achebe
Hardware-in-the-Loop Rig	laboratory, two rigs, one of which lags the current sensor configuration	the harness changes with each firmware release	5	one region	Nathan Oduya

3. Deployment topology and failure domains, beginning with Battery Management Service - part 2 of 2

Following on directly, these further rows of the 3. Deployment topology and failure domains, beginning with Battery Management Service belong to the same schedule and share its numbering. The parts are to be read together.

Scope, cut-off and units for the 3. Deployment topology and failure domains, beginning with Battery Management Service were fixed at its first part and are not repeated, being unaltered. The sequence is unbroken either side of it.

Schedule B - Deployment topology and failure domain by service (TDR-2026-13) (continued, part 2 of 2)

Service	Deployment topology	Release cadence	Instances per region	Failure domain	Engineering owner
Fleet Health Monitor	cloud, regional	fortnightly	2	one region	Wanda Njoku
Battery Management Service	split edge and cloud	pack firmware twice a year, the cloud service monthly	3	one robot	Peter Halvorsen
Gripper Controller	edge, only on the two robot variants fitted with a manipulator	quarterly	6	one robot	Ines Duarte
Map Authoring Studio	cloud, single global deployment	weekly	1	one region	Lucia Ferrante
Site Provisioning Service	cloud, global	fortnightly	3	one region	Marcus Duplessis
Event Bus	cloud, one cluster per region, three availability zones	upgraded twice a year	6	one region	Erik Palmgren
Config Distribution	split edge and cloud, with a signed bundle format	weekly	3	one robot	Sam Oyelaran
Robot Shadow Store	cloud, regional, sharded by site	weekly	4	one region	Marta Kellerman
Analytics Warehouse Loader	cloud, one loader per region writing to a shared warehouse	weekly	3	one region	Farida Cisse
Billing Usage Meter	cloud, regional	monthly, aligned to the billing cycle	5	one region	Nadia Semenova
Alarm Router	cloud, regional, with a second delivery provider	fortnightly	5	one region	Wanda Njoku
Video Snippet Service	cloud, regional, with an on-robot capture buffer	monthly	6	one region	Anjali Sekhon
Speed and Separation Monitor	edge, cross-checked against the safety controller's own inputs	twice a year, with revalidation	3	one robot	Bettina Vogel
Wheel Odometry Filter	edge	with the firmware, twice a year	3	one robot	Yuki Harada
Payload Scale Interface	edge, only on robots fitted with the weighing deck	as needed	4	one robot	Ines Duarte
Edge Gateway Agent	edge, one per robot, with mutual authentication to the	every three weeks	6	one robot	Sam Oyelaran

Service	Deployment topology	Release cadence	Instances per region	Failure domain	Engineering owner
	cloud				
Release Orchestrator	cloud, single global deployment	continuous, self-hosting	1	one region	Colin Achebe

4. Interface boundaries and data ownership around Robot Shadow Store

Where Robot Shadow Store is concerned, the commitment is that shadow state is no more than two seconds behind the robot at p95; the reviewer traced the metric to its source. Robot Shadow Store is deployed cloud, regional, sharded by site. Robot Shadow Store was assessed on the shadow consistency note, staleness measurements and a rebalance observation, and nothing beyond that was available in the data room. Release cadence for Robot Shadow Store is weekly, and the release records bear that out. The Robot Shadow Store codebase is Go 1.22 throughout, deployed onto cloud, containers with a sharded cache, and its durable record is Redis 7 as the hot path with PostgreSQL 14 as the durable record. Robot Shadow Store is the clearest instance in TDR-2026-13 of a control that exists on paper and is weaker in practice.

Robot Shadow Store was checked for shared-database coupling, and the ownership boundary held. The fleet allocator, the traffic manager and the remote operations console rely on Robot Shadow Store, which in turn relies on the telemetry ingest path and the edge gateway agent. The backlog for Robot Shadow Store is 9 items deep and contains an acknowledged defect whereby a shard rebalance briefly serves stale shadow state without marking it stale. Automated test coverage on Robot Shadow Store stands at 68 per cent line coverage. Robot Shadow Store was weighed on its coverage, its defect backlog and its dependency position, and no verified defect was identified, and two plausible concerns are recorded below at low evidence strength. Robot Shadow Store is not material on its own and belongs in this report because of what it reveals about the surrounding process.

Ownership of Localization Engine sits with Autonomy Core; Rahul Menon signs off changes to it. The cadence recorded for Localization Engine is every three weeks with the planner, which matches the observed release history. Localization Engine was examined for unsupported components; an unmaintained scan-matching library is still linked in the fallback path. Test coverage on Localization Engine is 66 per cent line coverage and the measurement basis was confirmed with Rahul Menon. The evidence base for Localization Engine was the localization design note, survey reports from three sites and the reviewer's map audit. Localization Engine illustrates why TDR-2026-13 separates the severity of a finding from the confidence behind it.

The topology for Localization Engine is edge, with map distribution from the cloud, and the reviewer confirmed it against the deployment manifests. 16 defects remain open against Localization Engine; the one the reviewer pressed on is that relocalisation after a lift-and-place takes longer than the operator prompt allows. The lidar driver, the wheel odometry filter and the map distribution channel feed Localization Engine; a failure in Localization Engine is felt by the motion planner and the fleet health monitor. The objective recorded against Localization Engine is that pose error stays under 25 mm in mapped aisles at survey speed, and it is reported weekly to the engineering leadership. Localization Engine was checked for shared-database coupling, and the ownership boundary held. Localization Engine was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

Release cadence for Map Authoring Studio is weekly, and the release records bear that out. The interface Map Authoring Studio publishes is versioned, and the map distribution channel and the localization engine were checked for use of any undocumented field. 22 items sit in the Map Authoring Studio backlog; the entry that bears on the investment case is that an unpublished draft can be exported and hand-loaded, bypassing the review step. Map Authoring Studio runs cloud, single global deployment; that placement is what determines whether a fault is one robot or one region. Map Authoring Studio reports 49 per cent line coverage, measured by the same instrumentation the pipeline uses. Map Authoring Studio carries a follow-up request that was satisfied after the draft of TDR-2026-13 was circulated.

The Map Authoring Studio codebase is TypeScript with a Python service layer throughout, deployed onto cloud, a desktop-class web application, and its durable record is PostgreSQL 14 with map revisions in object storage. Evidence examined for Map Authoring Studio comprised the authoring workflow document, the revision history for six sites and an export audit. Upstream of Map Authoring Studio sit the site survey imports and the identity service; downstream, the map distribution channel and the localization engine consume its output. The reviewer's judgement on Map Authoring Studio is that the position is stronger than the coverage figure alone would suggest, because the bench evidence is unusually

good here. The ageing position for Map Authoring Studio is a geometry rendering library one major version behind its supported line, and the reviewer checked it against the vendor statements on file. Map Authoring Studio was one of two items where TDR-2026-13 could not obtain the underlying data within the engagement.

Simulation Harness runs cloud batch, with a local mode for developers; that placement is what determines whether a fault is one robot or one region. Simulation Harness sits at 47 per cent line coverage; the trend across the reviewed period was up by three points. On end-of-life exposure, Simulation Harness shows the physics core depends on a library branch that stopped receiving fixes in 2023. Simulation Harness publishes a stable interface, though two fields used by the release orchestrator gate and the autonomy regression report are documented only in code comments. The dependency map places the planner and localization builds under test above Simulation Harness and the release orchestrator gate and the autonomy regression report below it. Simulation Harness is the clearest instance in TDR-2026-13 of engineering practice being better than the documentation suggests.

On paper Simulation Harness releases continuous, from the main branch, and the observed practice did not diverge from that. Colin Achebe of Developer Productivity maintains Simulation Harness, and the service inventory entry was current at the review date. What the reviewer had for Simulation Harness was the scenario catalogue, two weeks of nightly run records and a determinism experiment; the conclusions below do not reach past it. Simulation Harness pairs a Python 3.11 with a C++ physics core implementation with developer workstations and a batch cluster, and its data lives in object storage for scenario definitions and run artefacts. Against Simulation Harness the backlog runs to 31 items, and it includes a known behaviour whereby scenario determinism breaks when the physics core is rebuilt with a different optimisation level. Simulation Harness illustrates why TDR-2026-13 separates the severity of a finding from the confidence behind it.

For Wheel Odometry Filter the stated target is that odometry drift stays under 1.5 per cent of distance travelled on a clean floor, expressed against production traffic rather than a benchmark. Read against its objective, Wheel Odometry Filter the design is sound and the test evidence is thinner than the design deserves. Wheel Odometry Filter shows 62 per cent line coverage in the pipeline, and the number is credible and the measurement basis is honest. Wheel Odometry Filter ships with the firmware, twice a year; the reviewer sampled the last twelve releases. On end-of-life exposure, Wheel Odometry Filter shows none identified. Wheel Odometry Filter was inspected in the repository rather than described from the ticket, which is why the account is specific.

Wheel Odometry Filter carries 8 open defects, among them a case where drift exceeds the specification on wet or dusty floors and is not flagged to the operator. Underlying Wheel Odometry Filter in this report are the filter design note, drift measurements at four sites and a wet-floor bench run. In production Wheel Odometry Filter is edge, unchanged since the last architecture review. Wheel Odometry Filter depends on the wheel encoders and the inertial measurement unit and is depended on by the localization engine and the motion planner. Yuki Harada leads Robot Firmware's work on Wheel Odometry Filter and was the reviewer's principal interview for it. Wheel Odometry Filter was reviewed against the position twelve months earlier, and the direction of travel is favourable.

Schedule C - Producer and consumer interface contracts examined (TDR-2026-13)

Producer service	Consumer	Interface form	Interface version	Compatibility check	Breaking changes
Motion Planner	traffic manager	internal protocol	v2.8	enforced in the pipeline	1 in the reviewed period
Motion Planner	on-robot drive supervisor	request-response interface	v2.7	enforced in the pipeline	1 in the reviewed period
Costmap Fusion	motion planner	published schema	v3.9	not automated	none
Costmap Fusion	speed	streamed topic	v2.10	enforced in the pipeline	2 in the reviewed period
Fleet Allocator	traffic manager	request-response interface	v4.10	not automated	2 in the reviewed period
Fleet Allocator	fleet health monitor	shared event contract	v1.7	enforced in the pipeline	1 in the reviewed period
Traffic Manager	motion planner	shared event contract	v3.4	advisory only	1 in the reviewed period
Traffic Manager	remote operations console	streamed topic	v3.4	enforced in the pipeline	none
Charge Scheduler	fleet allocator	published schema	v2.6	not automated	2 in the reviewed period
Charge Scheduler	fleet health monitor	request-response	v1.8	not automated	2 in the reviewed

Producer service	Consumer	Interface form	Interface version	Compatibility check	Breaking changes
		interface			period
Safety Controller	drive power contactors	published schema	v2.5	advisory only	none
Safety Controller	robot fault annunciator	streamed topic	v1.2	enforced on the producer side	1 in the reviewed period
E-Stop Chain	safety controller	internal protocol	v4.9	not automated	1 in the reviewed period
E-Stop Chain	drive contactors	shared event contract	v4.5	not automated	1 in the reviewed period
Lidar Driver	costmap fusion pipeline	streamed topic	v4.8	advisory only	1 in the reviewed period
Lidar Driver	localization engine	published schema	v3.1	advisory only	2 in the reviewed period
Localization Engine	motion planner	streamed topic	v2.0	enforced on the producer side	2 in the reviewed period
Localization Engine	fleet health monitor	published schema	v2.7	not automated	2 in the reviewed period
WMS Connector	order broker	streamed topic	v4.1	not automated	2 in the reviewed period
WMS Connector	billing usage meter	shared event contract	v1.2	not automated	2 in the reviewed period
Order Broker	fleet allocator	request-response interface	v1.4	enforced in the pipeline	none
Order Broker	analytics warehouse loader	published schema	v4.11	enforced on the producer side	2 in the reviewed period
Telemetry Ingest	fleet health monitor	internal protocol	v3.8	enforced in the pipeline	1 in the reviewed period
Telemetry Ingest	alarm router	request-response interface	v2.7	enforced in the pipeline	none
OTA Updater	on-robot supervisor	shared event contract	v1.7	not automated	2 in the reviewed period
OTA Updater	fleet health monitor	streamed topic	v2.10	enforced on the producer side	2 in the reviewed period
Identity and Authorization	every cloud service	shared event contract	v4.10	enforced on the producer side	2 in the reviewed period
Identity and Authorization	remote operations console	published schema	v1.8	advisory only	2 in the reviewed period
Remote Operations Console	traffic manager	request-response interface	v2.11	not automated	none
Remote Operations Console	alarm router	published schema	v3.0	enforced on the producer side	1 in the reviewed period
Simulation Harness	release orchestrator gate	request-response interface	v4.10	enforced on the producer side	2 in the reviewed period
Simulation Harness	autonomy regression report	request-response interface	v2.4	not automated	none
Hardware-in-the-Loop Rig	firmware release decision record	published schema	v2.3	enforced on the producer side	1 in the reviewed period

4. Interface boundaries and data ownership around Robot Shadow Store - part 2 of 2

Presentation of the 4. Interface boundaries and data ownership around Robot Shadow Store carries on below without restarting the row numbers, the break being one of pagination only. Nothing is restated from the earlier part.

No basis of compilation changes at this break; whatever the 4. Interface boundaries and data ownership around Robot Shadow Store recorded about scope and method at its head continues to apply. No row is omitted at the join.

Schedule C - Producer and consumer interface contracts examined (TDR-2026-13) (continued, part 2 of 2)

Producer service	Consumer	Interface form	Interface version	Compatibility check	Breaking changes
Fleet Health Monitor	alarm router	streamed topic	v4.10	enforced on the producer side	none
Fleet Health Monitor	field service workflow	published schema	v4.6	advisory only	2 in the reviewed period
Battery Management Service	charge scheduler	request-response interface	v4.11	not automated	1 in the reviewed period
Battery Management Service	fleet health monitor	streamed topic	v2.1	enforced on the producer side	2 in the reviewed period
Gripper Controller	order execution path	published schema	v2.4	enforced in the pipeline	1 in the reviewed period
Gripper Controller	safety controller fault	published schema	v3.5	enforced on the	2 in the reviewed

Producer service	Consumer	Interface form	Interface version	Compatibility check	Breaking changes
	input			producer side	period
Map Authoring Studio	map distribution channel	internal protocol	v3.4	advisory only	1 in the reviewed period
Map Authoring Studio	localization engine	internal protocol	v1.8	not automated	none
Site Provisioning Service	config distribution service	published schema	v1.6	enforced on the producer side	none
Site Provisioning Service	WMS connector	internal protocol	v2.10	enforced in the pipeline	1 in the reviewed period
Event Bus	every consuming service	published schema	v1.5	advisory only	none
Config Distribution	on-robot supervisor	streamed topic	v2.1	enforced on the producer side	1 in the reviewed period
Config Distribution	motion planner parameters	internal protocol	v1.3	enforced on the producer side	2 in the reviewed period
Robot Shadow Store	fleet allocator	shared event contract	v4.9	enforced on the producer side	2 in the reviewed period
Robot Shadow Store	remote operations console	streamed topic	v1.9	enforced on the producer side	2 in the reviewed period
Analytics Warehouse Loader	customer reporting views	streamed topic	v4.2	enforced in the pipeline	1 in the reviewed period
Analytics Warehouse Loader	internal metrics layer	shared event contract	v2.1	enforced on the producer side	2 in the reviewed period
Billing Usage Meter	finance export	streamed topic	v2.7	enforced in the pipeline	1 in the reviewed period
Billing Usage Meter	customer usage report	streamed topic	v3.8	advisory only	none
Alarm Router	paging provider	streamed topic	v2.3	not automated	none
Alarm Router	customer notification channel	request-response interface	v3.6	not automated	1 in the reviewed period
Video Snippet Service	remote operations console	shared event contract	v2.5	enforced in the pipeline	none
Video Snippet Service	incident review workflow	shared event contract	v3.9	enforced in the pipeline	none
Speed and Separation Monitor	safety controller	streamed topic	v1.3	enforced in the pipeline	none
Speed and Separation Monitor	motion planner speed limit	published schema	v1.6	advisory only	none
Wheel Odometry Filter	localization engine	request-response interface	v1.2	not automated	1 in the reviewed period
Wheel Odometry Filter	motion planner	published schema	v2.7	enforced on the producer side	1 in the reviewed period
Payload Scale Interface	gripper controller	request-response interface	v2.0	enforced on the producer side	none
Payload Scale Interface	order execution path	shared event contract	v4.9	not automated	2 in the reviewed period
Edge Gateway Agent	telemetry ingest path	streamed topic	v4.3	enforced in the pipeline	2 in the reviewed period
Edge Gateway Agent	robot shadow store	streamed topic	v2.1	advisory only	2 in the reviewed period
Release Orchestrator	OTA updater	request-response interface	v1.4	enforced in the pipeline	none
Release Orchestrator	cloud deployment controllers	internal protocol	v4.0	advisory only	none

5. Automated test coverage across the reviewed services

Interface discipline around E-Stop Chain is adequate; the schema is versioned and change is gated on a compatibility check. VHDL with discrete relay logic is the implementation language for E-Stop Chain, which is hosted on a safety-rated hardware assembly, no general-purpose operating system with no persistent store; events are latched to the safety controller behind it. E-Stop Chain was weighed on its coverage, its defect backlog and its dependency position, and the component is adequate today and carries a supportability liability that is not on the funded plan. The manifests place E-Stop Chain edge, hardwired, independent of the compute module. E-Stop Chain sits between the operator stop buttons and the perimeter interlocks and the safety controller and the drive contactors, and that position is what makes its availability objective load-bearing. E-Stop Chain would be a candidate for the first post-close engineering review under any reasonable plan.

Changes to Hardware-in-the-Loop Rig reach production the harness changes with each firmware release. For Hardware-in-the-Loop Rig the supportability position is the rig controller runs an operating system release that left vendor support in 2025. What the reviewer had for Hardware-in-the-Loop Rig was the bench suite definition, rig configuration inventory and two firmware release records; the conclusions below do not reach past it. The firmware candidate artefacts and the bench sensor stimulators feed Hardware-in-the-Loop Rig; a failure in Hardware-in-the-Loop Rig is felt by the firmware release decision record. Hardware-in-the-Loop Rig was assessed against the objective and the defect record, and the component is adequate today and carries a supportability liability that is not on the funded plan. Hardware-in-the-Loop Rig was inspected in the repository rather than described from the ticket, which is why the account is specific.

The manifests place Motion Planner edge, one instance per robot with no cloud fallback. 71 per cent line coverage is reported for Motion Planner, and the reviewer re-ran the measurement on a clean checkout. Priya Raghunathan of Autonomy Core maintains Motion Planner, and the service inventory entry was current at the review date. The published objective for Motion Planner is that plan cycle completes inside 40 ms at the 99th percentile. What the reviewer had for Motion Planner was eleven design records, the planner benchmark suite and four weeks of production plan traces; the conclusions below do not reach past it. Motion Planner is the clearest instance in TDR-2026-13 of engineering practice being better than the documentation suggests.

Hosting for Speed and Separation Monitor is on-robot Linux, pinned to the safety compute path, and its C++17 with a safety-reviewed core code writes through to a rolling in-memory window with fault snapshots to the safety controller. Upstream of Speed and Separation Monitor sit the costmap fusion pipeline and the safety-rated scanner inputs; downstream, the safety controller and the motion planner speed limit consume its output. On Speed and Separation Monitor the conclusion is qualified: it is fit for the current fleet size, and the reviewer found no evidence that it would remain so at twice that scale. Speed and Separation Monitor is promoted twice a year, with revalidation, with the ring order fixed by policy. For Speed and Separation Monitor the reviewer worked from the separation monitoring design note, the internal safety case and six bench approach tests. Speed and Separation Monitor is not material on its own and belongs in this report because of what it reveals about the surrounding process.

Schedule D - Automated coverage trend on the reviewed services (TDR-2026-13)

Service	Coverage 2025-09	Coverage 2026-03	Coverage at review	Change over period	Owning team
Motion Planner	63%	70%	71%	+8	Autonomy Core
Costmap Fusion	51%	55%	58%	+7	Perception
Fleet Allocator	74%	79%	82%	+8	Fleet Services
Traffic Manager	65%	72%	74%	+9	Fleet Services
Charge Scheduler	54%	60%	63%	+9	Fleet Services
Safety Controller	82%	88%	91%	+9	Safety Engineering
E-Stop Chain	n/a	n/a	witnessed test	not comparable	Safety Engineering
Lidar Driver	37%	44%	44%	+7	Robot Firmware
Localization Engine	61%	66%	66%	+5	Autonomy Core
WMS Connector	60%	68%	69%	+9	Integrations
Order Broker	72%	75%	78%	+6	Fleet Services
Telemetry Ingest	55%	58%	61%	+6	Data and Telemetry
OTA Updater	67%	73%	76%	+9	Platform Infrastructure
Identity and Authorization	77%	82%	84%	+7	Security Engineering
Remote Operations Console	48%	51%	52%	+4	Applications
Simulation Harness	44%	46%	47%	+3	Developer Productivity
Hardware-in-the-Loop Rig	35%	36%	39%	+4	Robot Firmware
Fleet Health Monitor	50%	54%	57%	+7	Field Reliability
Battery Management Service	45%	51%	54%	+9	Robot Firmware
Gripper Controller	38%	38%	41%	+3	Robot Firmware
Map Authoring Studio	43%	48%	49%	+6	Applications
Site Provisioning Service	55%	62%	64%	+9	Platform Infrastructure
Event Bus	n/a	n/a	witnessed test	not comparable	Platform

Service	Coverage 2025-09	Coverage 2026-03	Coverage at review	Change over period	Owning team
					Infrastructure
Config Distribution	68%	69%	72%	+4	Platform Infrastructure
Robot Shadow Store	63%	68%	68%	+5	Fleet Services
Analytics Warehouse Loader	42%	43%	45%	+3	Data and Telemetry
Billing Usage Meter	75%	77%	79%	+4	Data and Telemetry
Alarm Router	46%	50%	51%	+5	Field Reliability
Video Snippet Service	37%	43%	43%	+6	Applications
Speed and Separation Monitor	82%	87%	87%	+5	Safety Engineering
Wheel Odometry Filter	53%	60%	62%	+9	Robot Firmware
Payload Scale Interface	29%	33%	36%	+7	Robot Firmware
Edge Gateway Agent	67%	73%	74%	+7	Platform Infrastructure
Release Orchestrator	47%	53%	55%	+8	Developer Productivity

6. What the incident record shows about the architecture

Lidar Driver owns timing event INC-2025-047, which the reviewer classes as timing. Evidence strength on the incident INC-2025-047 analysis is moderate on the cause and strong on the delivery position, which was checked in the repository. Remediation for INC-2025-047 is to re-derive the time offset on every reconnect and add the affected variant to the second bench rig; Yuki Harada owns it and the committed date is 2025-06-15. The driver fix landed on 2025-04-05; the second rig was still unmodified at review is the position on the Asterion Logistics incident INC-2025-047 commitment as at the review date. 2025-04-02 is the date of timing incident INC-2025-047, on which a lidar unit reconnected after a cable seating fault and reported frames with a stale timestamp offset. Asterion Logistics incident INC-2025-047 has no single accountable owner, and that is the finding the reviewer would press hardest.

INC-2025-047 was caused by a design gap: the driver did not re-derive the sensor time offset on reconnect. In INC-2025-047, the bench rig that would have caught it carries a different lidar variant, which is a process finding rather than a code finding. Taken with the rest of the record, Asterion Logistics incident INC-2025-047 indicates that management's severity rating is defensible and its remediation priority is not. The blast radius of timing incident INC-2025-047 was one site and 12 robots. Attention came to incident INC-2025-047 when an automated fusion timing alert fired inside four minutes. incident INC-2025-047 was cross-checked against the engineering record supplied for TDR-2026-13, and the two agree.

2026-01-13 is the date of Localization Engine incident INC-2026-004, on which relocalisation after a lift-and-place exceeded the operator prompt timeout at a newly surveyed site. The commitment register shows Localization Engine incident INC-2026-004 against an undertaking to replace the fallback matcher and re-survey the affected aisles, with Rahul Menon named and 2026-06-30 committed. Reported detection applies to Localization Engine incident INC-2026-004, since the site's commissioning engineer raised it during acceptance testing. Localization Engine owns performance incident INC-2026-004, which the reviewer classes as performance. The engineering team's root cause for Kestrel Medical Supply incident INC-2026-004 is that the fallback scan matcher is slower than the primary and is selected whenever the map cache misses, and the reviewer found the analysis sound. INC-2026-004 was compared with the equivalent position at two other components, and it is neither the best nor the worst.

Grading Kestrel Medical Supply incident INC-2026-004 on evidence rather than on consequence gives limited, being a single account corroborated only by the ticket timeline. Performance incident INC-2026-004 carries a second-order finding: the site was surveyed with a sparse feature set that increases cache misses. Performance incident INC-2026-004 affected one site and 18 robots. Verification of the incident INC-2026-004 remediation gives not landed; the matcher replacement is in design and the re-survey is scheduled. The two durations recorded for Localization Engine incident INC-2026-004 are not a service outage; 5 days to a workaround to mitigate and open at the review date to resolve. performance event INC-2026-004 is reported here in full because a summary of it in an earlier draft of TDR-2026-13 was misread as an assurance.

In taxonomy terms INC-2025-139 is a capacity incident in Event Bus. 2 h 47 m elapsed before capacity incident INC-2025-139 was mitigated, with resolution at 63 days. So far as reach is concerned, incident INC-2025-139 involved 7 sites and 246 robots. How INC-2025-139 surfaced matters to the assessment: the traffic manager's own latency alert fired first. Evidence strength on the incident INC-2025-139 analysis is moderate on the cause and strong on the delivery position, which was checked in the repository. multiple sites incident INC-2025-139 is not material on its own and belongs in this report because of what it reveals about the surrounding process.

The reviewer treats one contributing factor in incident INC-2025-139 as the more important lesson: the topic inventory had recorded the sharing as acceptable in 2023 and it was never revisited. Erik Palmgren committed after Event Bus incident INC-2025-139 to split control traffic onto a dedicated topic with its own quota and revisit the topic inventory quarterly, due 2026-01-31. Reading multiple sites incident INC-2025-139 from its first entry, on 2025-11-02 a telemetry backlog on a shared topic delayed control events by up to eleven seconds. Root cause in capacity incident INC-2025-139 was that control and telemetry traffic shared one topic and one consumer group budget. For incident INC-2025-139 the follow-through was tested rather than assumed, and the split landed on 2026-01-04; the quarterly review has not yet run. capacity incident INC-2025-139 illustrates why TDR-2026-13 separates the severity of a finding from the confidence behind it.

Underneath Crestline Fulfillment incident INC-2025-041 lies the design decision that the reservation lease had no expiry, so release depended on the departing robot. The reviewer treats one contributing factor in Crestline Fulfillment incident INC-2025-041 as the more important lesson: the alert rule for queue depth existed but was scoped to a metric the site did not emit. INC-2025-041 records that on 2025-03-14, a robot lost its network link mid-traverse and its corridor reservation was never released. Classifying resource leak event INC-2025-041 as resource leak matters because the Traffic Manager backlog contains two further items of the same kind. INC-2025-041 closure status is landed on 2025-05-22, verified in the reservation replay suite. Traffic Manager incident INC-2025-041 is treated as material to the investment case rather than as a routine engineering item.

The two durations recorded for incident INC-2025-041 are 1 h 52 m to mitigate and 9 days to resolve. In resource leak event INC-2025-041 the first signal was a site supervisor telephoned the on-call engineer after robots queued at one intersection, and the reviewer treats that as reported detection. What resource leak incident INC-2025-041 tells the reviewer is that the elapsed resolution time reflects a scheduling decision rather than an engineering difficulty. The reviewer grades the resource leak incident INC-2025-041 evidence as moderate on the cause and strong on the delivery position, which was checked in the repository and has not upgraded the finding to compensate. So far as reach is concerned, incident INC-2025-041 involved 2 sites and 61 robots. incident INC-2025-041 was reviewed against the position twelve months earlier, and nothing measurable has changed.

Stale state event INC-2025-078 was caused by a design gap: the session list cached the site roster and only invalidated it on a page reload. Evidence strength on the stale state incident INC-2025-078 analysis is strong, resting on the production traces the team supplied. INC-2025-078 was contained to 4 site or tenant with no robot taken out of service. Drawing the evidence in stale state incident INC-2025-078 together, the fix is real and the regression protection is not, so recurrence remains possible. Stale state event INC-2025-078 produced one commitment: to invalidate the roster cache on a fleet membership event and add a session-level regression test, owned by Anjali Sekhon, dated 2025-08-31. incident INC-2025-078 was checked against the company's own register, which had not been updated since the change landed.

2025-06-24 is the date of INC-2025-078, on which the console listed robots that had been removed from a site and an operator commanded a decommissioned unit. Beyond the direct cause, INC-2025-078 shows that no test covered a roster change during an open operator session. Remote Operations Console owns Harborline Systems incident INC-2025-078, which the reviewer classes as stale state. On delivery, stale state incident INC-2025-078 shows this: landed on 2025-07-28; the regression test was still marked flaky at review. Stale state event INC-2025-078 was detected because an internal engineer noticed the discrepancy during a demonstration. Remote Operations Console incident INC-2025-078 was reviewed against the position twelve months earlier, and the direction of travel is favourable.

Detection came through the robot self-reported the fault and the safety case review picked it up, which places hardware fault event INC-2025-090 on the reported side of the ledger. Safety Controller owns hardware fault event INC-2025-090, which the reviewer classes as hardware fault. Immediate stop, 22 m to return to service elapsed before

INC-2025-090 was mitigated, with resolution at 41 days. Incident INC-2025-090 affected one site and 3 robots. On delivery, Fathom Mobility incident INC-2025-090 shows this: the inspection change landed on 2025-08-04; nine units await retrofit. Safety Controller incident INC-2025-090 sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

The commitment register shows INC-2025-090 against an undertaking to add the connector torque check to incoming inspection and retrofit the affected harness batch, with Bettina Vogel named and 2025-09-30 committed. Underneath INC-2025-090 lies the design decision that a marginal connector on one channel produced intermittent readings inside the accepted noise band. Evidence strength on the INC-2025-090 analysis is strong, resting on the production traces the team supplied. The contributing factor recorded against INC-2025-090 is that the connector torque specification was not part of the incoming inspection sheet. INC-2025-090 is not read as an isolated failure: the remediation addresses the immediate cause and leaves the contributing factor in place. INC-2025-090 carries a follow-up request that was satisfied after the draft of TDR-2026-13 was circulated.

The two durations recorded for INC-2025-146 are not applicable; caught before billing to mitigate and 11 days to resolve. The classification applied to process defect event INC-2025-146 is process defect, and the affected component is Billing Usage Meter. In process defect event INC-2025-146 the first signal was the monthly reconciliation caught it before invoicing, and the reviewer treats that as reported detection. Process defect event INC-2025-146 would have been shorter or smaller had it not been that the runbook described the reset in prose rather than making it part of the command. INC-2025-146 affected no production site and no robot directly, the exposure being to internal systems. incident INC-2025-146 was raised again with the owning engineer after the initial interview, and the position did not change.

2025-11-19 is the date of Larkspur Home incident INC-2025-146, on which a re-run of a failed metering window double-counted usage for one customer. What went wrong in process defect incident INC-2025-146 was that the re-run procedure required a manual ledger reset that the operator skipped. The judgement recorded against process defect incident INC-2025-146 is that the remediation addresses the immediate cause and leaves the contributing factor in place. The reviewer checked the INC-2025-146 commitment against the code and release records; landed on 2025-11-30. Process defect incident INC-2025-146 produced one commitment: to fold the reset into the re-run command and make double-counting detectable in the ledger, owned by Nadia Semenova, dated 2025-12-31. process defect incident INC-2025-146 would be a candidate for the first post-close engineering review under any reasonable plan.

7. Third-party components inside the reviewed services

Cell-model-lib sits inside Battery Management Service and is therefore in the shipped product rather than in tooling. What the company must actually do about cell-model-lib is to keep the dynamic-linking boundary intact. Cell-model-lib carries Apache-2.0 terms, classified no for copyleft purposes. cell-model-lib illustrates why TDR-2026-13 separates the severity of a finding from the confidence behind it.

Usage of sqlite-embedded 3.45.1 is on-robot local state, and the inventory reference is OSS-010. Licensing for sqlite-embedded is public domain dedication and the copyleft column reads no. The reviewer's note on sqlite-embedded is that no attribution obligation. sqlite-embedded rests on a single source, and the reviewer has said so rather than implying corroboration.

Where jwt-verify came from matters here: public registry, signature verified. The non-licence risk on jwt-verify is a build-time dependency whose absence would stop the pipeline rather than the product. Because jwt-verify is used token verification in the identity service, its licence terms reach the delivered software. jwt-verify was checked against the company's own register, which had not been updated since the change landed.

Under MIT, csv-export is treated as no copyleft in the inventory. Provenance of csv-export is public registry, signature verified. Csv-export is recorded as approved 2023-10-17 in the component approval log. csv-export would be a candidate for the first post-close engineering review under any reasonable plan.

In addition to its terms, image-signing brings no identified risk beyond ordinary version currency. Approved 2025-01-15 is the approval entry for image-signing. Image-signing affects OTA Updater; no other component reviewed depends on it. image-signing was cross-checked against the engineering record supplied for TDR-2026-13, and the two agree.

Where opencv-subset came from matters here: public registry, signature verified. Opencv-subset is licensed under Apache-2.0, with copyleft exposure recorded as no. Opencv-subset affects Gripper Controller; no other component reviewed depends on it. opencv-subset is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Physics-core-fork is recorded as approved with conditions 2023-04-18 in the component approval log. Physics-core-fork obliges the company to share alike any derived asset it distributes. Licence text and the actual usage of physics-core-fork were read together, and the fork diverges by nineteen patches and no upstream merge path is maintained. physics-core-fork rests on a single source, and the reviewer has said so rather than implying corroboration.

What the company must actually do about http-router is to exclude a particular build configuration from the shipped artefact. Http-router entered the codebase by this route: public registry, signature verified. Http-router version 1.9.4 is catalogued at OSS-022 and is used cloud service routing. http-router was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

8. Architectural debt and the cost of removing it

As at the review date TD-10 was committed, with 2026-12-31 still the stated target. 0.5 engineer-quarters is the estimate on support item TD-10, which converts to $54,000. Funding for register item TD-10 is unfunded and dependent on capital rather than engineering effort, and the register does not say so. TD-10 was tested rather than accepted, and the test is described in the working papers behind TDR-2026-13.

Safety evidence item TD-18 is funded only if the team's headcount plan is met in the engineering plan the reviewer was shown. About programme item TD-18 it should be said that the safety team supports it; it is not funded in the current plan. TD-18 on Safety Controller cannot start earlier than it does because it is behind a platform change that has its own target date. TD-18 on Safety Controller was reviewed against the position twelve months earlier, and nothing measurable has changed.

TD-08 is funded from the second quarter onward; the register status and the funding position are not the same thing. Replace the archived charting library in the operator console is entered at TD-08 as an security supportability item. The reviewer's qualification on TD-08 on Remote Operations Console is that no advisory affects the current version, which is why it ranks low. programme item TD-08 is the clearest instance in TDR-2026-13 of a control that exists on paper and is weaker in practice.

Two templates remain forked as at the review date is the point the reviewer would put to management on TD-11 on Site Provisioning Service. Funding for TD-11 on Site Provisioning Service is not funded, and carried in the register regardless, and the register does not say so. Status on programme item TD-11 is in progress; the target is 2026-09-30. Site Provisioning Service item TD-11 is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Sizing on programme item TD-22 gives 0.25 engineer-quarters and $42,000. Funding for TD-22 on Payload Scale Interface is funded only if the team's headcount plan is met, and the register does not say so. Move the calibration record for the payload scale to the cloud is entered at quality item TD-22 as an quality item. Payload Scale Interface item TD-22 is the clearest instance in TDR-2026-13 of engineering practice being better than the documentation suggests.

The register shows TD-24 as committed, due 2026-09-30. TD-24 matters because it three modified files under a file-level copyleft licence appear unpublished. The reviewer's qualification on TD-24 is that counsel was engaged after the reviewer raised it. Costmap Fusion item TD-24 is the clearest instance in TDR-2026-13 of a control that exists on paper and is weaker in practice.

9. Where the architecture meets the safety case

Post-mitigation, Minor injury hazard HZ-14 sits at Tolerable once the stale roster defect is closed and is recorded as verified. Register entry HZ-14 scores Minor injury on severity; exposure is Occasional and controllability Low. For hazard HZ-14 in the register the verifying activity was console interaction test plus a review of session records. register entry HZ-14 is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

On the register's own scale, Serious injury hazard HZ-20 is Serious injury severity, Occasional exposure, Medium controllability. Occasional-exposure hazard HZ-20 produced one event whose investigation is still open; that is worth stating alongside the assessed exposure. Residual risk at hazard HZ-20 is stated as Tolerable where cleaning compliance is evidenced, and the status is partially verified. HZ-20 is not material on its own and belongs in this report because of what it reveals about the surrounding process.

After mitigation, register entry HZ-04 is Tolerable once calibration is gated in the service workflow; the register status is verified. Serious injury hazard HZ-04 has never activated in the field and is evidenced only on the bench, which is the strongest available test of the control. Reading HZ-04 against the verification evidence, the gate landed in March 2026 after a field incident. register entry HZ-04 rests on a single source, and the reviewer has said so rather than implying corroboration.

Register entry HZ-01 was verified through witnessed approach testing on six robots plus a review of the on-robot safety log. On Serious injury hazard HZ-01 the reviewer records that the fixed approach speed assumption is carried as an open qualification. Produced two near-miss reports from site staff is the incident history behind Serious injury hazard HZ-01. register entry HZ-01 was excluded from an earlier draft of TDR-2026-13 and reinstated after the reviewer obtained the supporting record.

What the register does not say about HZ-10, and the reviewer does, is that the supplier's certificate is held in the safety file. Against Fire hazard HZ-10 the team relies on contactor sequencing with an arc-suppression design in the charge dock. The hazard recorded at Improbable-exposure hazard HZ-10 is that charging contact arcs during an unplanned disconnect. Improbable-exposure hazard HZ-10 has no single accountable owner, and that is the finding the reviewer would press hardest.

Method of verification on HZ-11: supplier abuse test report plus a review of pack fault records. Reading Fire hazard HZ-11 against the verification evidence, state-of-health drift on opportunity-charged packs is an open modelling item. Field history for Improbable-exposure hazard HZ-11 is that it produced one field event that did not result in harm. hazard HZ-11 was one of two items where TDR-2026-13 could not obtain the underlying data within the engagement.

Hazard HZ-03 in the register is not clean: robots without the weighing deck rely on operator loading discipline. The control applied at HZ-03 is payload mass and centre-of-gravity limits enforced by the scale interface. Hazard HZ-03 in the register was scoped to every robot variant currently in production, and the reviewer checked the scoping against the fleet record. hazard HZ-03 was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

One caveat attaches to No direct harm hazard HZ-17: extraction runs weekly, so a busy robot can still lose events between extractions. The verification method for register entry HZ-17 is log extraction record review across the fleet. After mitigation, No direct harm hazard HZ-17 is Tolerable, with an evidence-loss caveat; the register status is open. Frequent-exposure hazard HZ-17 is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

10. Open items, declined remediations and qualifications

Brandywine Technology Partners will not treat an open item in this report as closed on a verbal assurance, and has said so to the engineering leadership. 5 items in report TDR-2026-13 are open, two of which the reviewer would treat as pre-closing information requests rather than post-close work. R. Ostrowski's conclusion in report TDR-2026-13 is that the engineering organisation is capable and under-instrumented, and that the risks identified are ordinary for a company at this stage rather than exceptional. On the evidence available by 2026-04-13, the present report concludes that the engineering organisation is capable and under-instrumented, and that the risks identified are ordinary for a company at this stage rather than exceptional. Cross-references in this report point to the evidence index rather than to a summary of the evidence.

Management's declinatures are reported in TDR-2026-13 at face value; the reviewer's disagreement is stated separately and does not overwrite them. The report at TDR-2026-13 treats a backlog label as evidence of intent and not as evidence of remediation, which is why several items remain open in this report that the company regards as handled. For Ardent Peak's purposes, the present report concludes that the engineering organisation is capable and under-instrumented, and that the risks identified are ordinary for a company at this stage rather than exceptional. Whatever weight a reader

gives the individual findings, report TDR-2026-13 concludes that the engineering organisation is capable and under-instrumented, and that the risks identified are ordinary for a company at this stage rather than exceptional. R. Ostrowski notes that the company answered every document request, though two answers arrived after the draft.

4 indicators are recommended in TDR-2026-13; the reviewer deliberately avoided proposing anything that needs a new pipeline. R. Ostrowski notes in TDR-2026-13 that the company's own reporting is adequate for three of the four recommended indicators. The open items carried forward from this report are set out below; a reader should treat the committed dates as intentions rather than facts. Three of the open items in this report would close on a single document that the company has agreed to produce. The company reviewed a draft of this report for factual accuracy and its corrections are reflected without altering any conclusion.

Where TDR-2026-13 says a defect is verified, the reviewer reproduced it or read a production trace of it; everything else is recorded as a concern. R. Ostrowski has left several concerns in this report at limited evidence strength rather than dropping them or dressing them up. TDR-2026-13 suggests one governance change rather than a price adjustment: a quarterly technical review with the lead investor's observer present. Brandywine Technology Partners notes that 2 of the matters in the present report are capable of being closed before completion if the company is asked in time. R. Ostrowski has stated the limitations at the point they bite rather than collecting them into a disclaimer.

11. Conclusions and implications for the transaction

The report at TDR-2026-13 concludes that the engineering organisation is capable and under-instrumented, and that the risks identified are ordinary for a company at this stage rather than exceptional. Stated without qualification for once, the conclusion of the present report is that the engineering organisation is capable and under-instrumented, and that the risks identified are ordinary for a company at this stage rather than exceptional. R. Ostrowski chose the sample for the present report after reading the dependency map, so the selection is not proportionate to headcount or to code volume. Sampling in the present report covered 10 of the 34 catalogued services and 4 of the 18 live sites. The service architecture, ownership boundaries and dependency structure of the fleet platform is a wide field, and the report at TDR-2026-13 is explicit about which parts of it were reached.

None of the findings in report TDR-2026-13 would in the reviewer's view justify withholding completion, and two would justify a specific undertaking. The costed remediation implied by this report is $1,605,000 over the coming four quarters, which is within the ordinary engineering budget. This report records 3 items the engineering team declined to remediate, each with the reason given and the reviewer's view of it. Two positions in the report at TDR-2026-13 are unresolved between the reviewer and the engineering team, and both are left unresolved on the page. This report contains 5 open items, and a reader should assume that number will move rather than settle.

R. Ostrowski would add one qualitative item to the pack proposed in the present report: the count of commitments closed without verifying evidence. The monitoring pack the report at TDR-2026-13 proposes would have surfaced two of the incidents examined here before a customer noticed. Taking the sections above together, the engineering organisation is capable and under-instrumented, and that the risks identified are ordinary for a company at this stage rather than exceptional. The judgement the present report reaches is that the engineering organisation is capable and under-instrumented, and that the risks identified are ordinary for a company at this stage rather than exceptional. Each section of the report at TDR-2026-13 states its own basis, so a reader can accept one section and reject another.